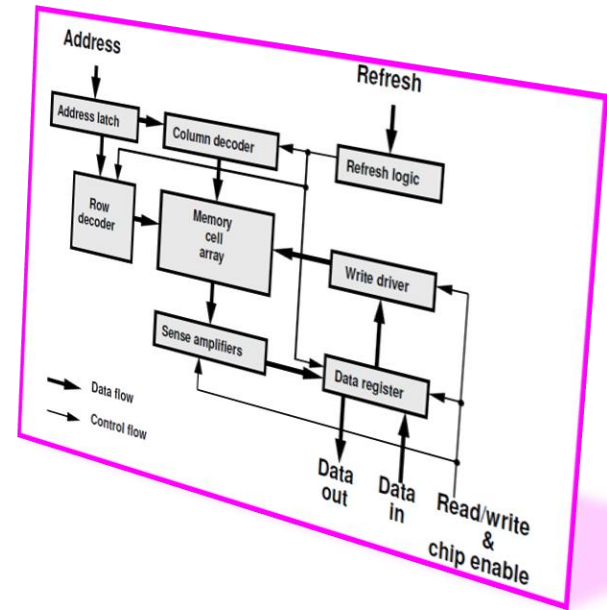


Outline

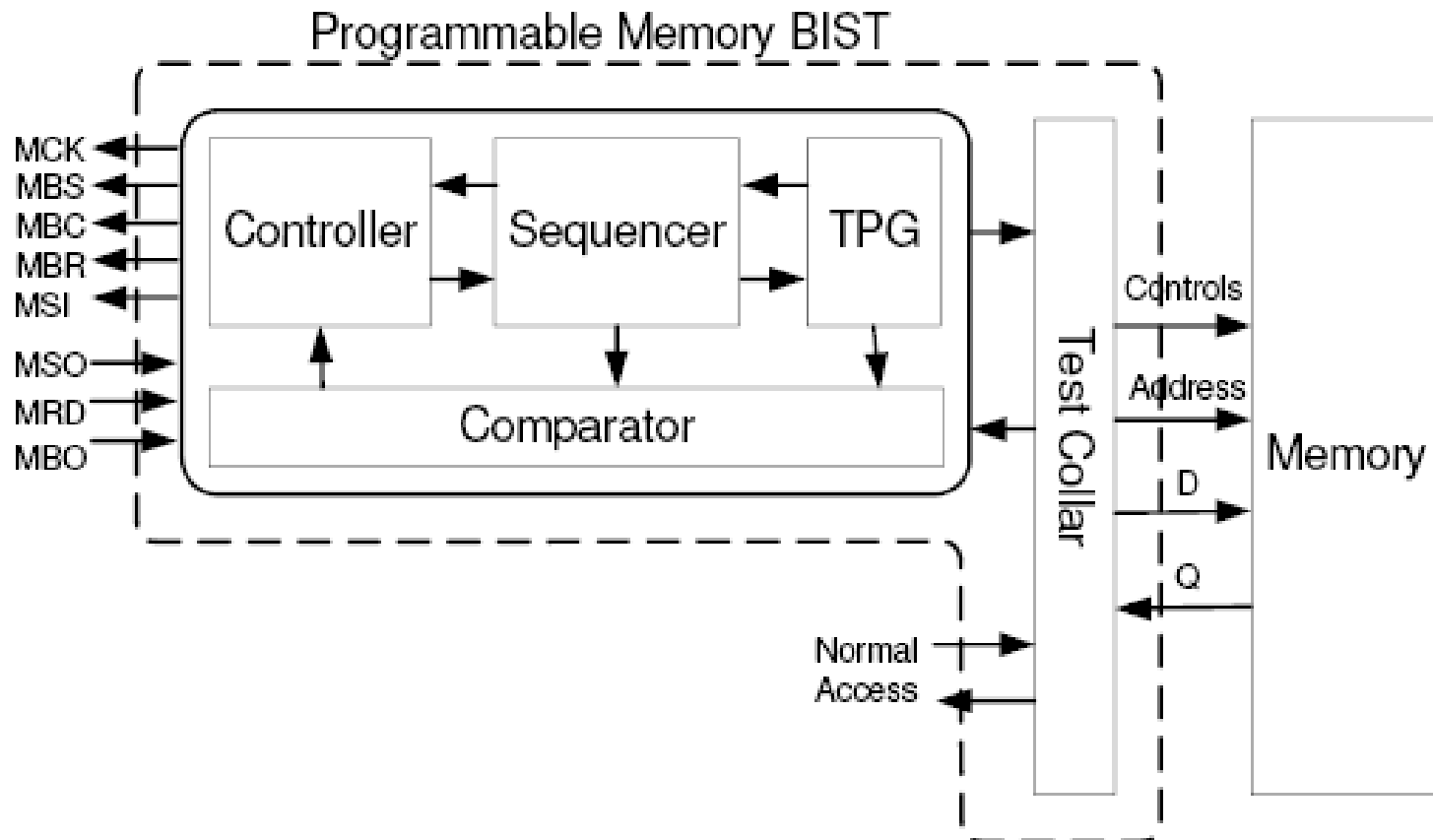
- Introduction
- Memory Fault Model
- Memory Test Algorithms
- Memory Fault Simulation* (not in exam)
- Memory Test Generation* (not in exam)
- Memory BIST* (not in exam)



Typical RAM BIST Approaches

- **Methodology**
 - ◆ **Processor-based BIST**
 - * **Programmable**
 - ◆ **Hardwired BIST**
 - * **Fast**
 - ◆ **Compact**
 - ◆ **Hybrid**
- **Interface**
 - ◆ **Serial (scan, 1149.1)**
 - ◆ **Parallel (embedded controller; hierarchical)**
- **Patterns (address sequence)**
 - ◆ **March & March-like**
 - ◆ **Pseudorandom**
 - ◆ **Others**

General RAM BIST Architectures

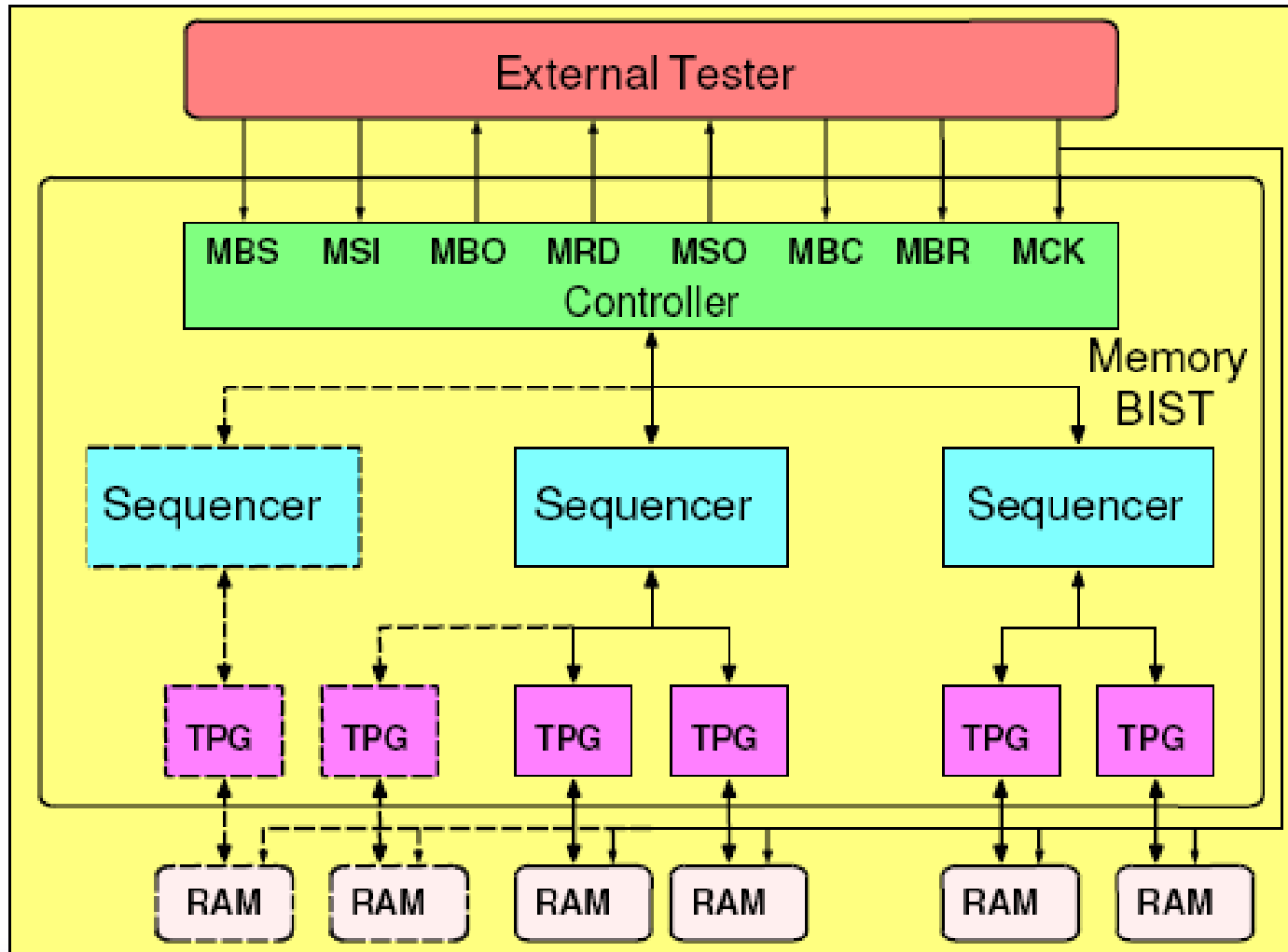


BIST I/O Pins

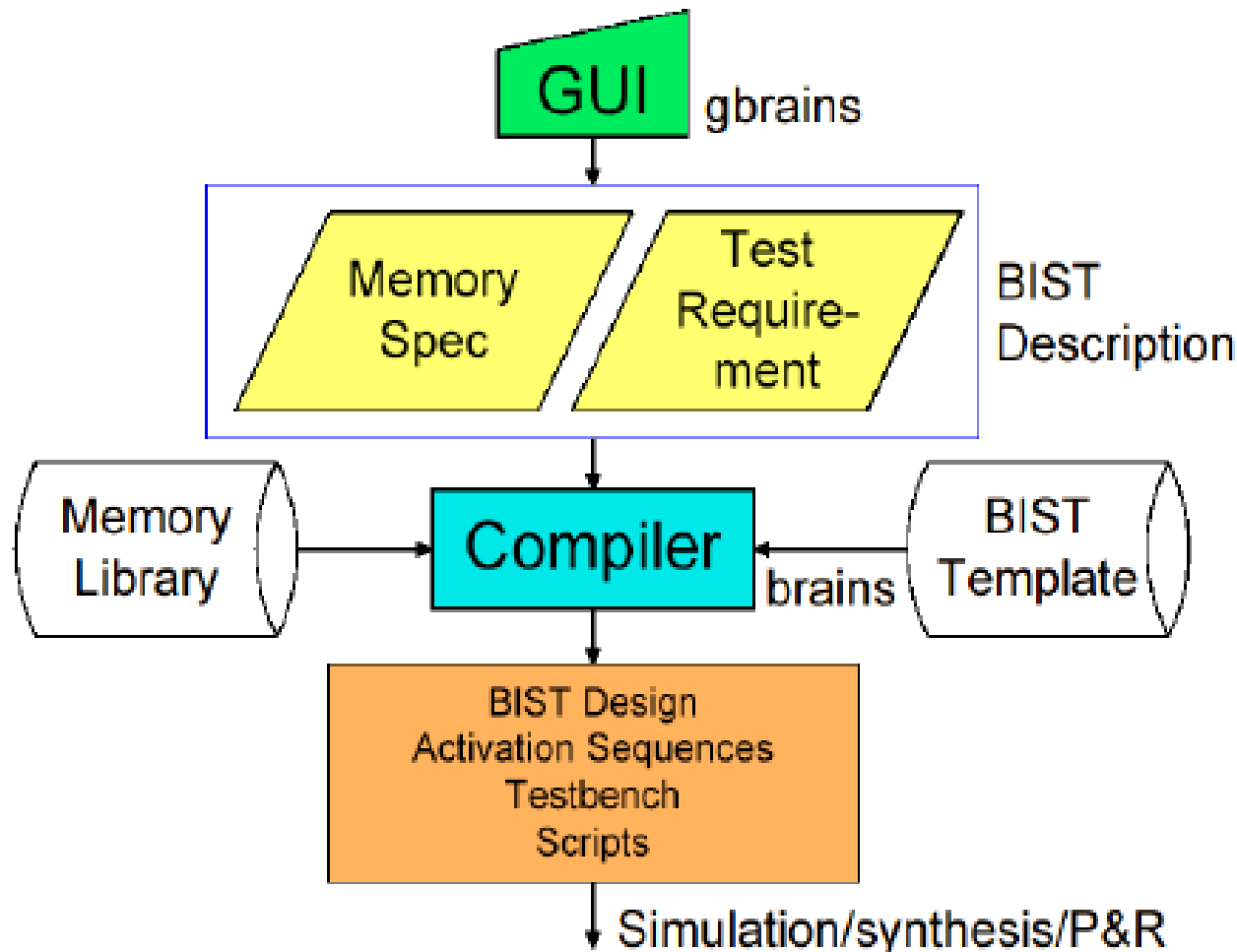
Name	IO	External IO	Descriptions
MBS	I	Yes	Memory BIST Selection
MBC	I	Yes	Memory BIST Control
MCK	I	Yes	Memory BIST Clock
MBR	I	Yes	Memory BIST Reset
MSI	I	Yes	Memory BIST command/data serial in
MSO	O	Yes	Memory BIST command/data serial out
MBO	O	Yes	Memory BIST Output
MRD	O	Yes	Memory BIST Output Ready
ADDR	O	No	Address Signals
D	O	No	Memory Data In
Q	I	No	Memory Data Out
CS	O	No	Chip Select
OE	O	No	Output Enable
WE	O	No	Write Enable

Multiple RAM Cores

- Controller and sequencer can be shared



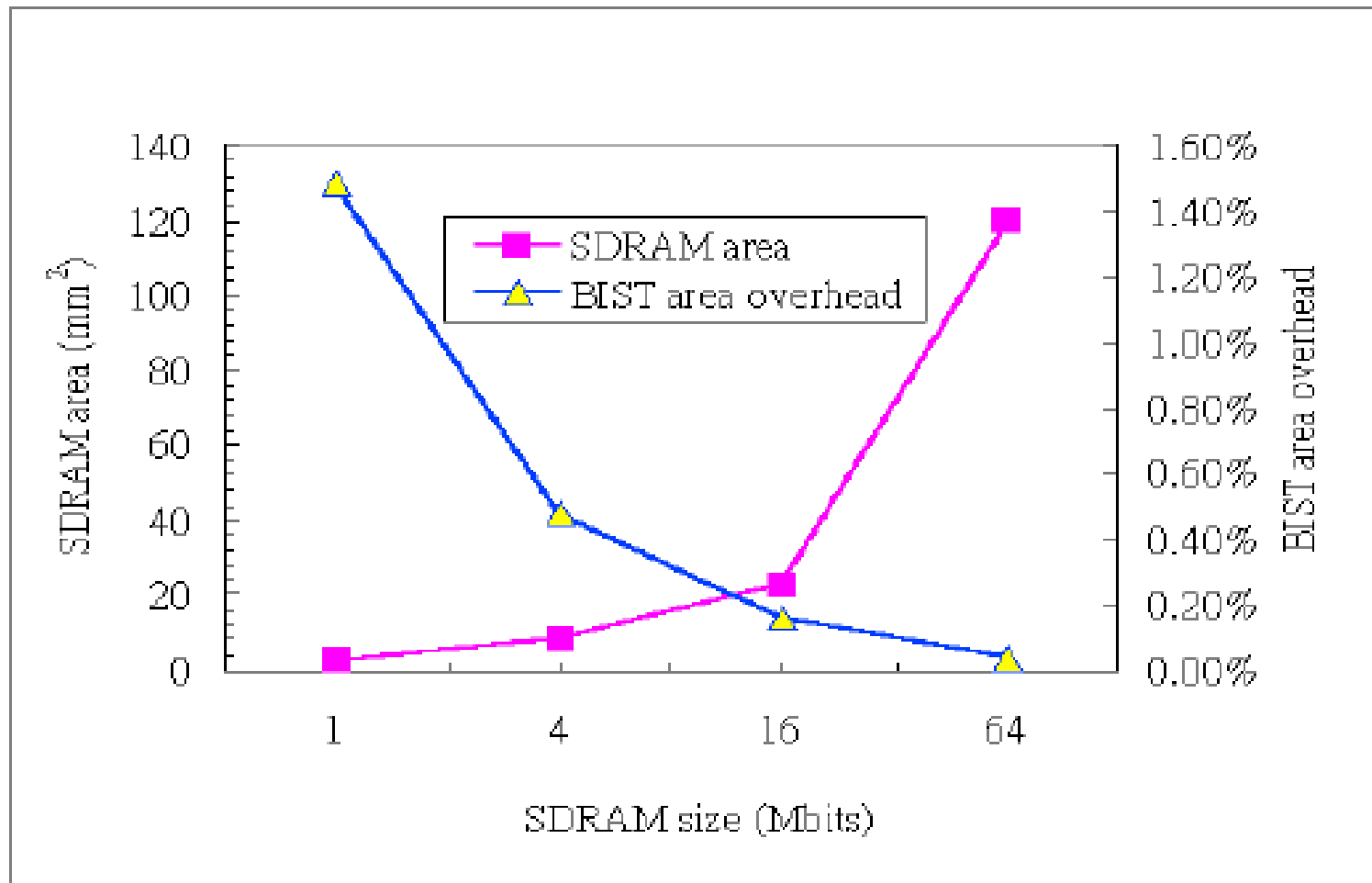
BRAINS Inputs and Outputs [Huang 99]



BRAINS Outputs

- **Synthesizable BIST design**
 - ◆ **At-speed testing**
 - ◆ **Programmable March algorithms**
 - ◆ **Optional diagnosis support**
 - * **Built-in Self Diagnosis, BISD**
- **Activation sequence**
 - ◆ **Test bench**
 - ◆ **Synthesis script**

Area Overhead



Summary

- **Memory testing are important**
- **Different memory requires different fault models**
- **March C is shortest march that detect all 4 fault models: $10N$**
- **RAM BIST are needed for embedded memories**
 - ♦ **EDA tools available**

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- [Nair 1979] R. Nair, “Comments on ‘An Optimal Algorithm for Testing Stuck-at Faults in Random Access Memory,’” *IEEE Trans. on Computers*, Vol. C-26, no. 11, pp.1141-1144, 1979.
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